PORTFOLIO

BAYU AJI TRISASONGKO

Electrical Engineering Graduate | Mechanical Design, Electronics, Mechatronics & Robotics

Bringing mechanical design, electronics, and control together to turn engineering ideas into tangible prototypes. An electrical engineering graduate with a solid hands-on foundation in mechatronics, component fabrication, and rapid prototyping. Eager to contribute and learn within engineering teams developing robotics, automation, and precision mechanical systems.

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Bayu Aji Trisasongko · S1 Teknik Elektro, Universitas Telkom



Telkom University
S1 Teknik Elektro (2026)

Mechanical
3D CAD modeling, cycloidal gearbox design, CNC & 3D print fabrication.
`SolidWorks` `Cycloidal Reducer` `FDM 3D Print`

Electronics
Schematic design, circuit wiring, PCB layout, and power electronics.
`KiCad/Altium` `Relay Control` `Power Distribution`

Embedded Control
STM32 & Arduino MCU firmware, magnetic sensing, and closed loop.
`STM32 C/C++` `AS5600 Encoder` `PID Closed-Loop`

Prototyping
Physical mechatronics build, functional bench testing, and validation.
`Hardware Bench` `Rapid Iteration` `System Validation`

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01 / PROJECTS

Engineering Work, in Practice · Selected Prototyping Showcase

Koleksi 5 proyek rekayasa fisik: Lengan Robot 5-DOF, Pendingin Surya, Robot Basket KRAI, Line Follower & Sumo Robot

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01 / FINAL PROJECT



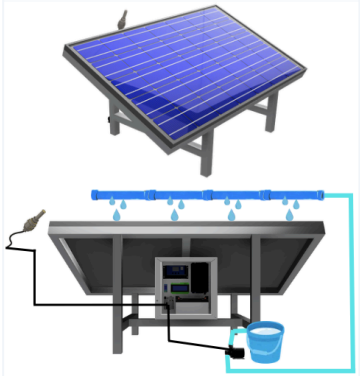
Robot Manipulator 5-DOF ↗

Prototipe lengan robot artikulasi 5-DOF bertenaga STM32F411 dengan gearbox cycloidal hasil 3D-print presisi. Mengintegrasikan encoder magnetik AS5600 closed-loop feedback untuk meminimalkan backlash pada sendi artikulasi gerak, mencapai akurasi kinematika 79,8% pada beban uji 250-1.000 g.

STM32F411 MCU	NEMA 17 Stepper
TB6600 Driver	AS5600 Sensor
TCA9548A Mux	Cycloidal Gearb…

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02 / CAPSTONE DESIGN



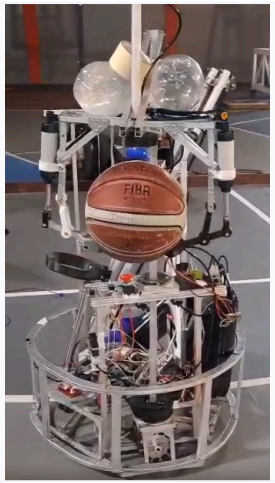
Pendingin Panel Surya Otomatis ↗

Sistem penyemprotan air otomatis untuk menurunkan temperatur operasional panel surya berbasis Arduino Uno. Memanfaatkan sensor suhu presisi dan relai pompa celup untuk memicu siklus pendinginan di atas >30°C dan berhenti di bawah <25°C demi memulihkan efisiensi daya optimal.

Arduino Uno R3	RTD Temp Sensor
5V Relay Module	12V Water Pump
16x2 I2C LCD	12V Battery Pack

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03 / KONTES ROBOT ABU INDONESIA (KRAI)



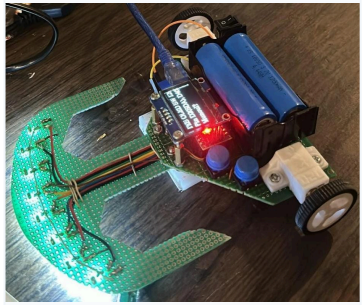
Kompetisi Robot Basket ↗

Robot penembak bola basket dinamis untuk Kontes Robot ABU Indonesia. Berperan aktif di tim mekanik merancang subsistem pelontar pneumatik 6-bar, mekanisme intake dribbler, serta chassis manuver roda omniwheel, berhasil lolos seleksi ke Tahap Nasional ITB Jatinangor.

6-Bar Pneumatics	24V Solenoid
RS775 DC Motor	Omniwheel Drive
Limit Switch & IR	Air Regulator

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04 / PROYEK MANDIRI



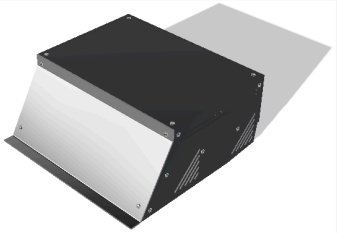
Robot Line Follower ↗

Robot pelacak garis otomatis berkecepatan tinggi dengan kemudi diferensial presisi. Dilengkapi bar sensor 8 pasang fotodioda LED, pemrosesan cepat mikrokontroler Arduino Nano, serta pengemudi motor TB6612FNG responsif dengan monitoring layar OLED 0.96 inci.

Photodiode Array	Arduino Nano
TB6612 Driver	N20 Gear Motor
0.96" I2C OLED	2S Li-ion Batte…

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05 / PROYEK MANDIRI



Robot Sumo ↗

Robot sumo kelas kompetisi 1 kg berdimensi 20×20 cm dengan sasis akrilik kokoh dan bilah sekop pelat aluminium. Mengandalkan 4 motor planetari PG24 torsi tinggi yang menghasilkan daya dorong 4 kg, dikendalikan nirkabel dari smartphone via Bluetooth HC-05.

Arduino Uno	4x PG24 Motors
L298N Driver	HC-05 Bluetooth
3S LiPo Battery	Acrylic Chassis

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5 Prototipe Rekayasa Fisik
Validasi hardware & fungsional 100%

Integrasi Mekatronika Lengkap
3D CAD, PCB, Firmware STM32 & Arduino

Pengalaman Kompetisi Robotik
Divisi Mekanik Nasional KRAI ITB

Dokumentasi & Video Lengkap
Model 3D interaktif di portobayu.vercel.app

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